# AI-Powered Resume Screening System

## Minor Project Report

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**Submitted By:**

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**Hall Ticket No: 23231A4222**

**B.Tech 4th Year**

**CSE (AIML)**

**Submitted To:**

**Pursuit Future Technologies Pvt. Ltd.**

**Domain:**

**Artificial Intelligence**

# INTR0DUCTION

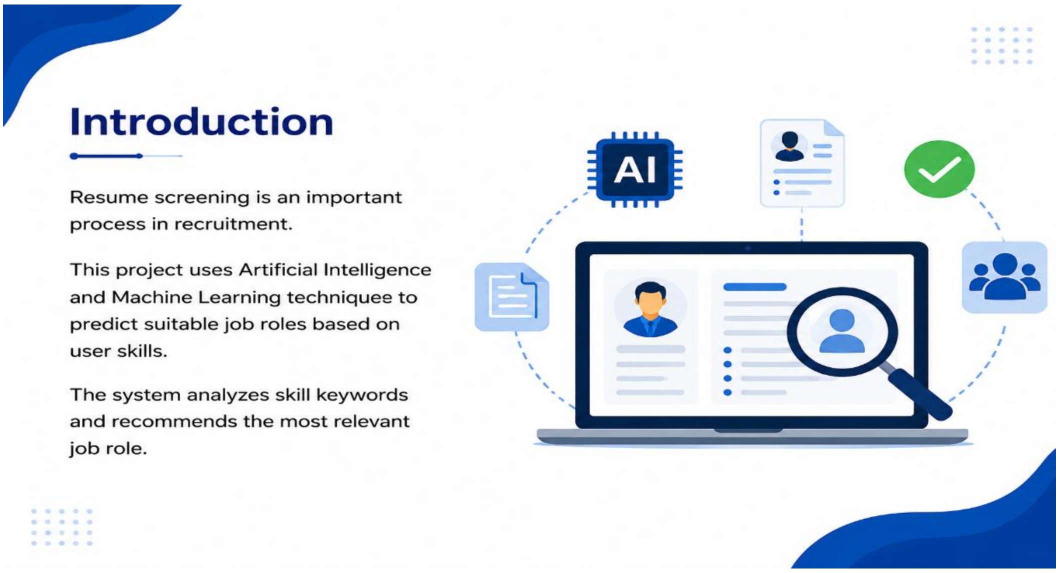
In today's digital world, recruitment and resume screening have become important processes for organizations and companies. Manually analyzing resumes requires a lot of time and effort, especially when there are many applicants for a single job role. To overcome this problem, Artificial Intelligence and Machine Learning technologies can be used to automate the resume screening process and provide faster and more accurate results.

The AI-Powered Resume Screening System is a simple Machine Learning-based application developed using Python and Streamlit. The main objective of this project is to predict suitable job roles based on the skills entered by the user. The system analyzes the entered skills and compares them with predefined job-related skill datasets to identify the most relevant job role.

This project uses Natural Language Processing (NLP) techniques and similarity matching algorithms for analyzing text data. The Count Vectorizer technique is used to convert textual skill data into numerical form, and Cosine Similarity is used to measure the similarity between the user input and stored skill data. Based on the similarity score, the system recommends the most appropriate job role to the user.

The project is developed with a simple and interactive user interface using the Streamlit framework, which allows users to easily enter their skills and receive predictions instantly. The application demonstrates the practical implementation of Machine Learning concepts in real-world recruitment systems.

This project helped in understanding Python programming, Machine Learning basics, NLP concepts, and user interface development. It also provided practical knowledge about how Artificial Intelligence can be used to simplify recruitment-related tasks and improve efficiency in job role prediction systems.



# PROJECT OBJECTIVES

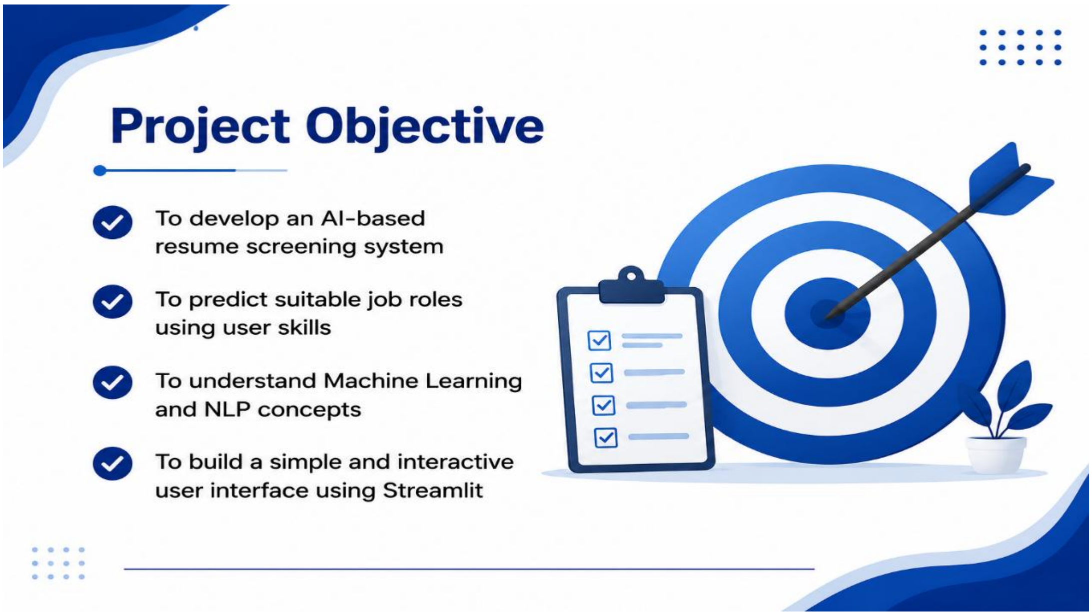
The main objective of this project is to develop an AI-based Resume Screening System that can predict suitable job roles based on the skills provided by the user. The project aims to simplify and automate the resume analysis process using Artificial Intelligence and Machine Learning techniques.

Another important objective of this project is to understand the practical implementation of Machine Learning concepts and Natural Language Processing techniques in real-world applications. The system processes textual skill data, converts it into numerical form, and compares it with predefined datasets to identify the most relevant job role.

The project is also designed to improve programming and development skills using Python and various Machine Learning libraries such as Pandas, Scikit-learn, and Streamlit. Through this project, knowledge about text processing, similarity matching, and user interface development is enhanced.

This project also aims to provide a simple, interactive, and user-friendly interface where users can easily enter their skills and receive instant predictions. The system demonstrates how AI technologies can be used to reduce manual effort, save time, and improve efficiency in recruitment and job recommendation systems.

Overall, the objective of this project is not only to build a functional AI application but also to gain practical exposure to Machine Learning workflows, NLP techniques, and modern Python-based application development.



# TECHNOLOGIES USED

The AI-Powered Resume Screening System is developed using various technologies, programming tools, and Machine Learning libraries that help in building an efficient and interactive application. **The primary programming language used in this project is Python**, which is widely used for Artificial Intelligence, Machine Learning, and data analysis applications because of its simplicity and powerful libraries.

**Streamlit :**The project uses the Streamlit framework to create a simple and user-friendly web interface. Streamlit helps in developing interactive applications quickly without requiring advanced web development knowledge. It allows users to enter their skills and instantly receive job role predictions through a clean graphical interface.

**Pandas library** is used for handling and processing the dataset. It helps in reading, organizing, and managing the resume skill data stored in CSV format. NumPy is also used for numerical operations and efficient data handling during processing.

**Scikit-learn** is one of the main Machine Learning libraries used in this project. It provides the Count Vectorizer technique, which converts textual skill data into numerical vectors that can be analyzed by the system. The project also uses the Cosine Similarity algorithm from Scikit-learn to compare the similarity between user-entered skills and predefined job-related skill datasets.

**Natural Language Processing (NLP)** concepts are used in this project to process and analyze textual information. NLP helps the system understand skill-related text data and perform accurate matching between different skill sets.

**Visual Studio Code (VS Code)** is used as the code editor for writing, editing, and executing the Python code. Additionally, CSV files are used for storing sample skill datasets and job role information in a simple and organized format.

Overall, these technologies collectively help in developing an efficient AI-based application that demonstrates the practical implementation of Machine Learning and NLP techniques in resume screening and job role prediction systems.

## SYSTEM WORKFLOW

**User Skills Input**

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**Text Processing**

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**Feature Extraction**

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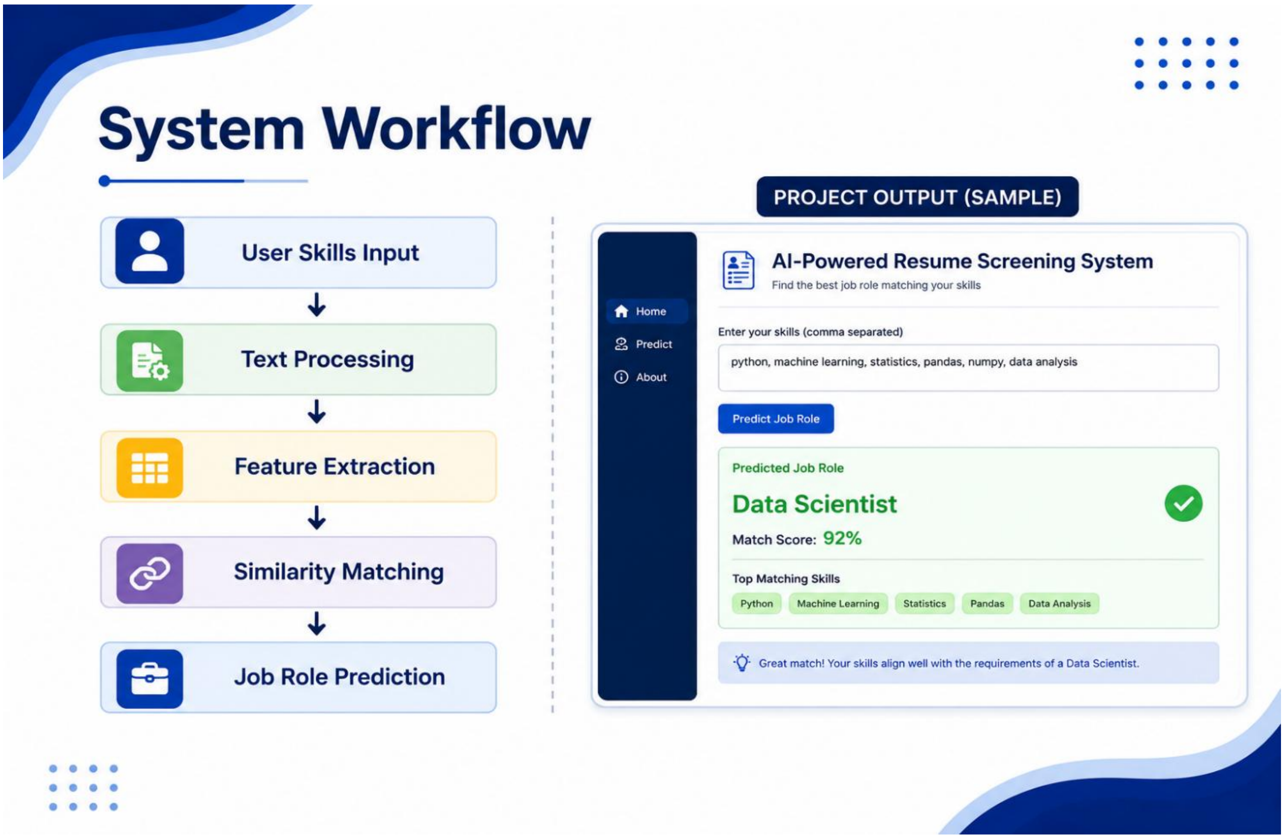
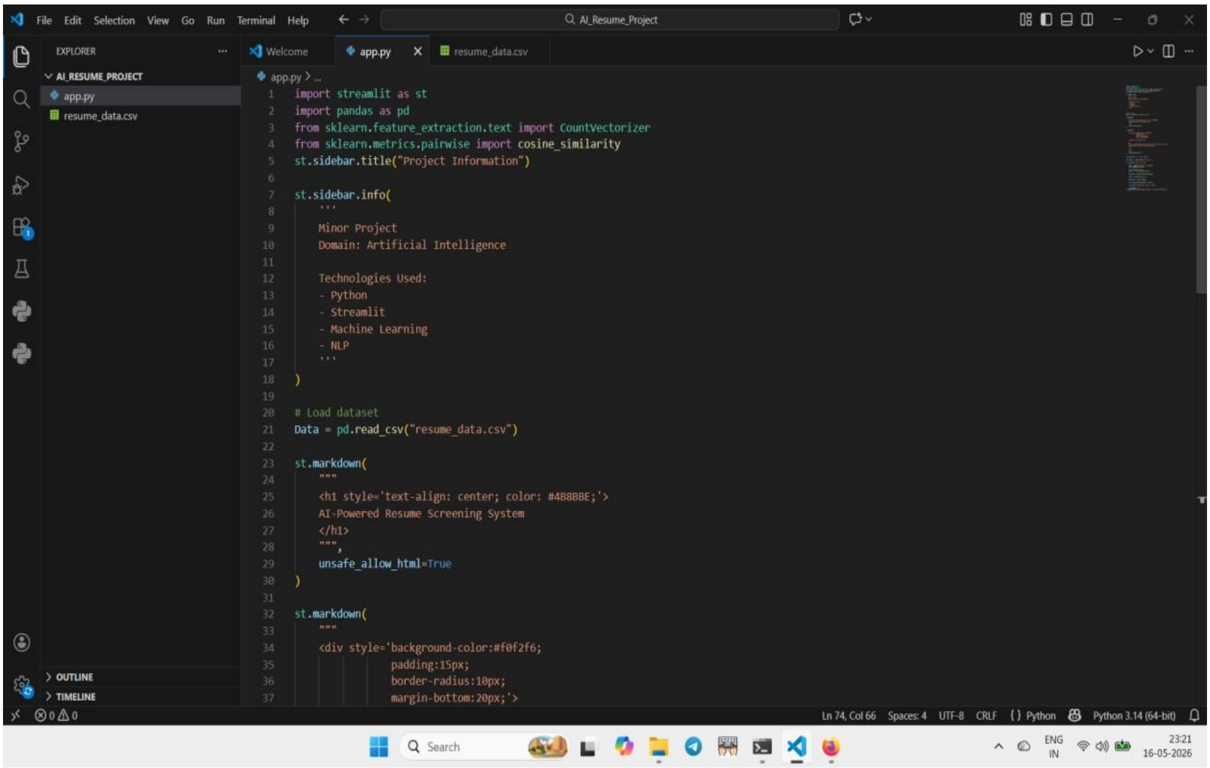
**Similarity Matching**

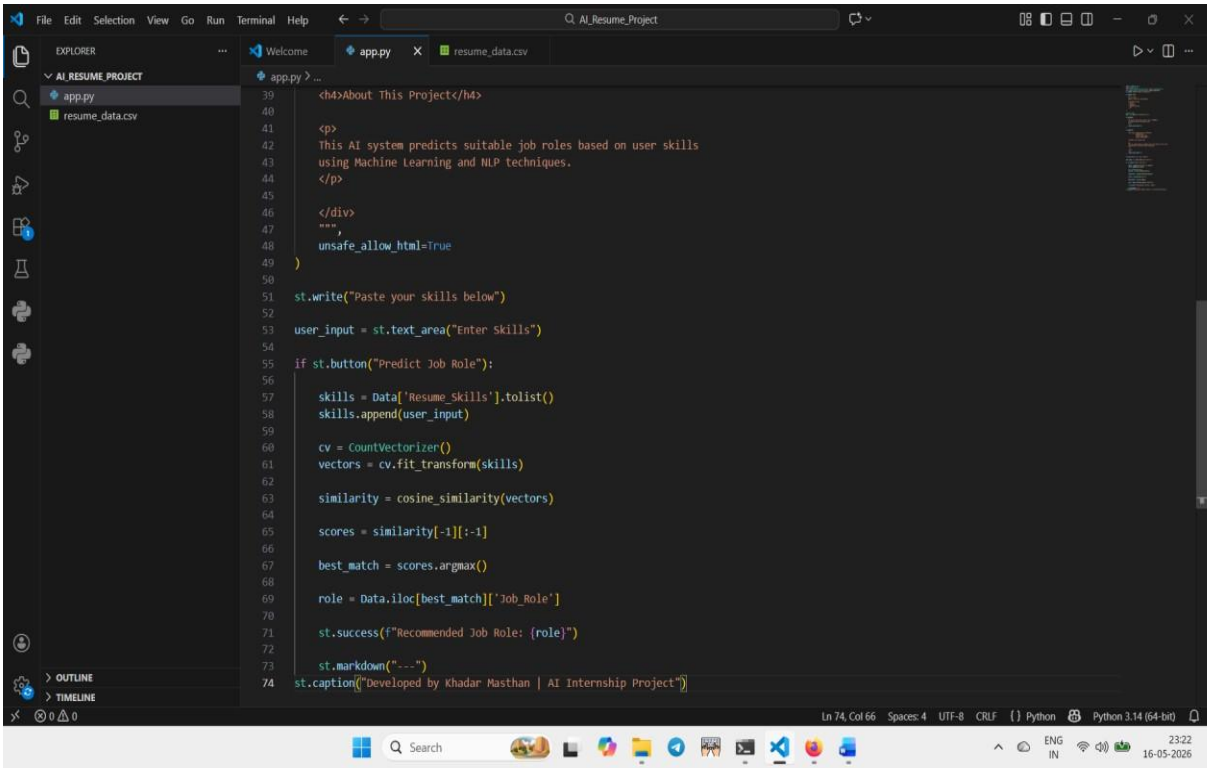
↓

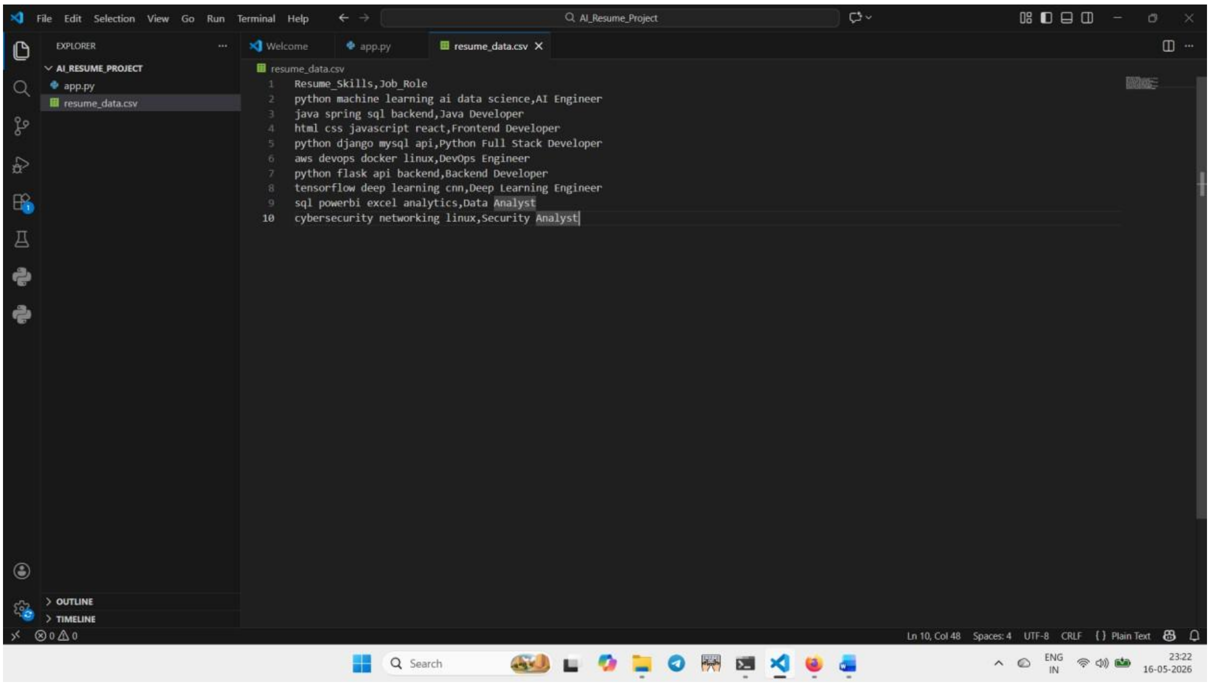
**Job Role Prediction**



# PROJECT OUTPUT

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AI_Resume_Project
EXPLORER
AI_RESUME_PROJECT
app.py
resume_data.csv
Welcome
app.py
resume_data.csv
Resume_Skills,Job_Role
python machine learning ai data science,AI Engineer
java spring sql backend,Java Developer
html css javascript react,Frontend Developer
python django mysql api,Python Full Stack Developer
aws devops docker linux,DevOps Engineer
python flask api backend,Backend Developer
tensorflow deep learning cnn,Deep Learning Engineer
sql powerbi excel analytics,Data Analyst
cybersecurity networking linux,Security Analyst
OUTLINE
TIMELINE
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ENG IN
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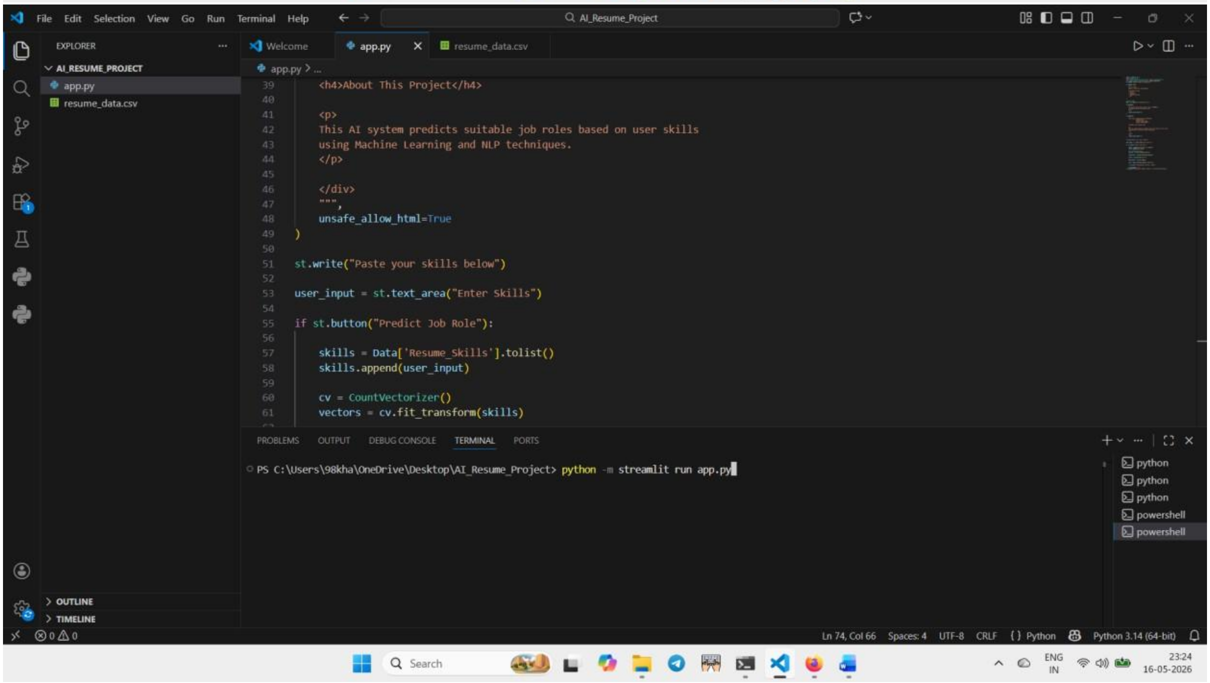
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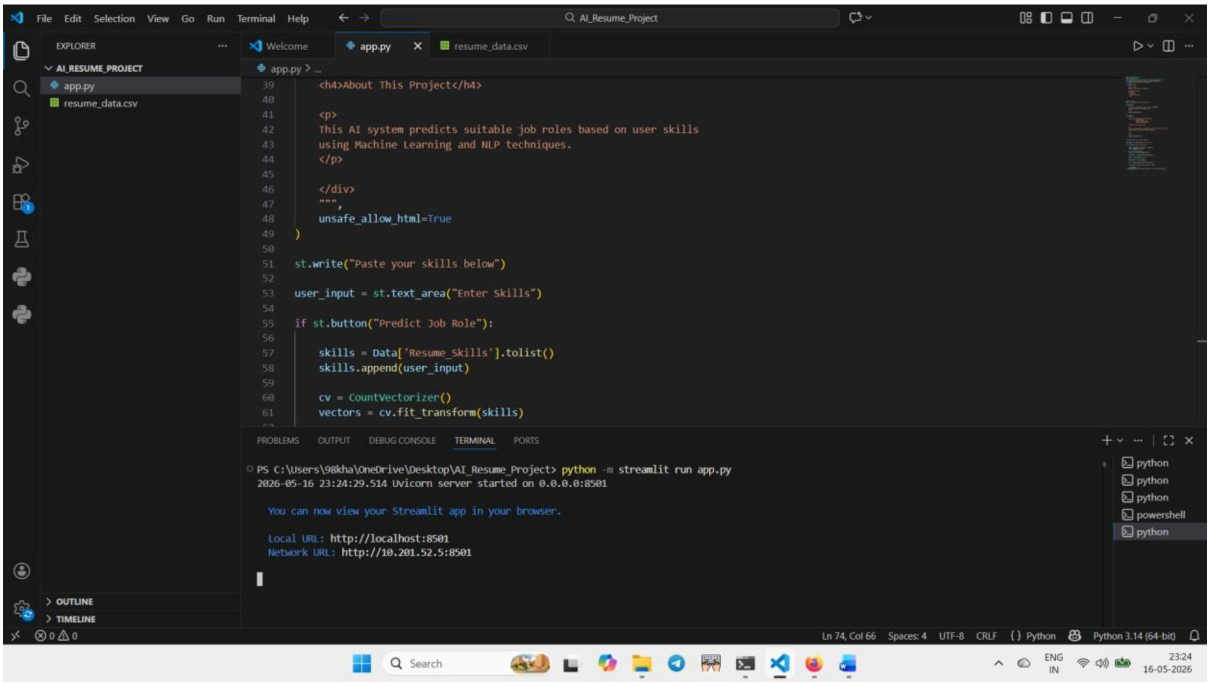
<h4>About This Project</h4>
<p>
This AI system predicts suitable job roles based on user skills
using Machine Learning and NLP techniques.
</p>
</div>
""",
unsafe_allow_html=True
)
st.write("Paste your skills below")
user_input = st.text_area("Enter Skills")
if st.button("Predict Job Role"):
skills = Data['Resume_Skills'].tolist()
skills.append(user_input)
cv = CountVectorizer()
vectors = cv.fit_transform(skills)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\98kha\OneDrive\Desktop\AI_Resume_Project> python -m streamlit run app.py
python
python
python
powershell
powershell
OUTLINE
TIMELINE
Ln 74, Col 66 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit)
Search
ENG IN
23:24
16-05-2026



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PS C:\Users\98kha\OneDrive\Desktop\AI_Resume_Project> python -m streamlit run app.py
2026-05-16 23:24:29.514 Uvicorn server started on 0.0.0.0:8501
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://10.201.52.5:8501
python
python
python
powershell
python
OUTLINE
TIMELINE
Ln 74, Col 66 Spaces: 4 UTF-8 CRLF {} Python Python 3.14 (64-bit)
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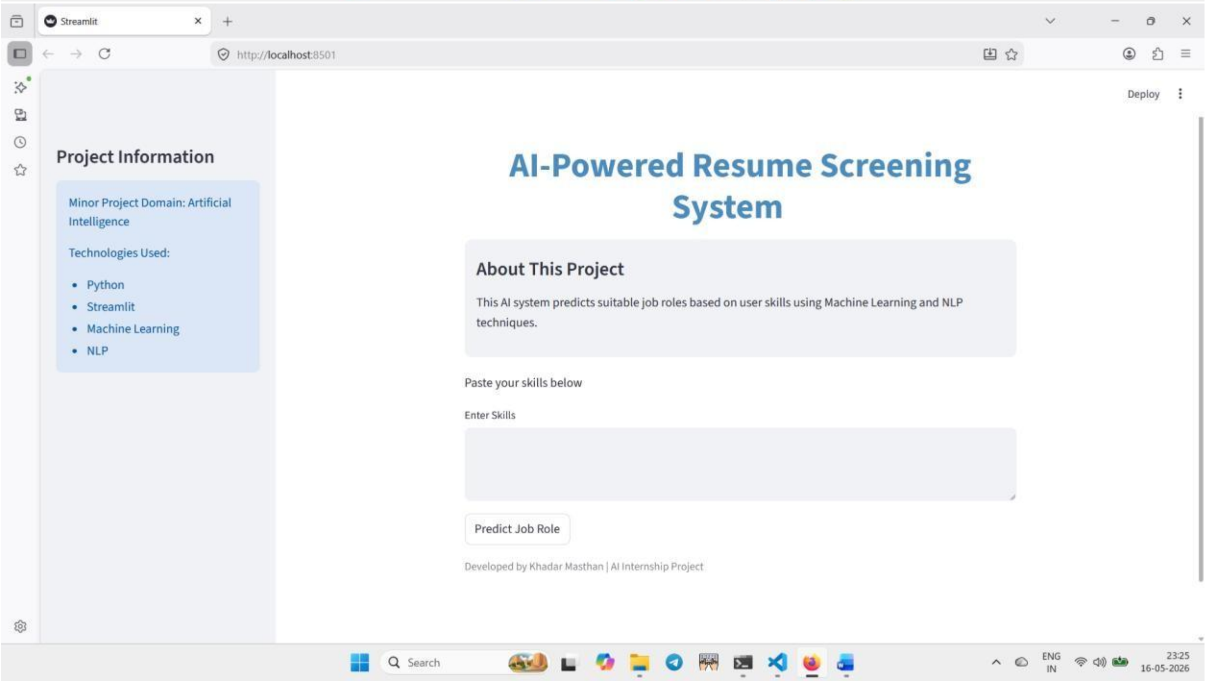

Streamlit
http://localhost:8501
Deploy
Project Information
Minor Project Domain: Artificial Intelligence
Technologies Used:
Python
Streamlit
Machine Learning
NLP
AI-Powered Resume Screening System
About This Project
This AI system predicts suitable job roles based on user skills using Machine Learning and NLP techniques.
Paste your skills below
Enter Skills
Predict Job Role
Developed by Khadar Masthan | AI Internship Project
Search
ENG IN
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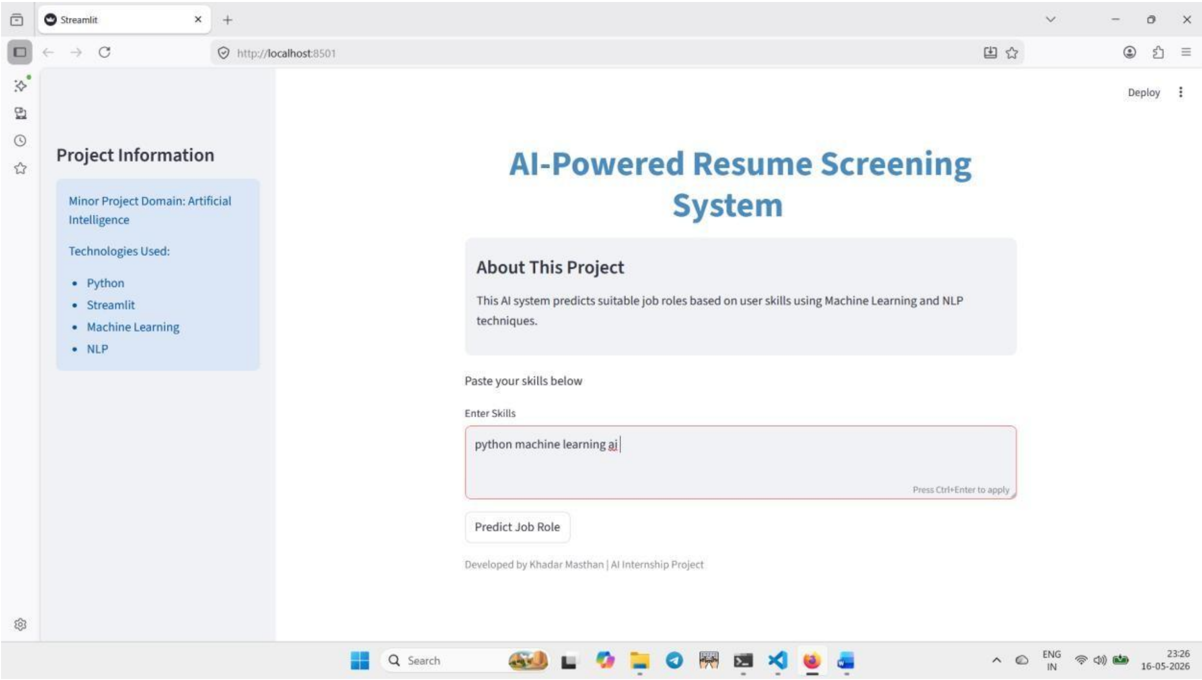
16-05-2026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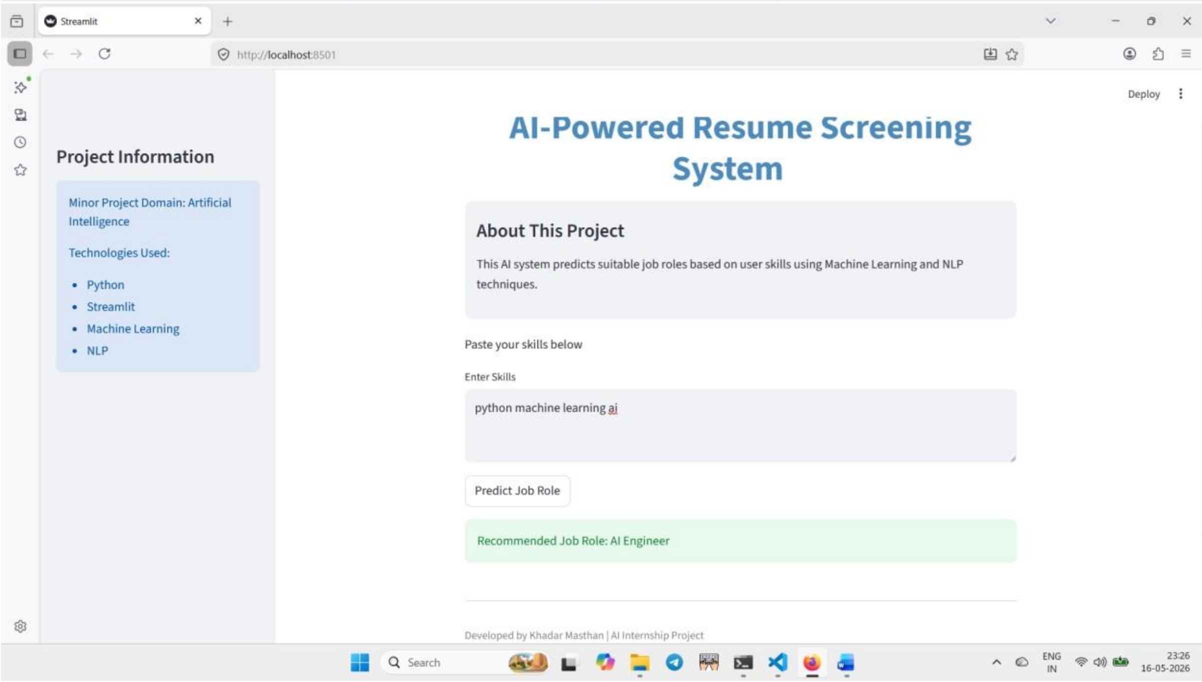
Paste your skills below
Enter Skills
python machine learning ai
Press Ctrl+Enter to apply
Predict Job Role
Developed by Khadar Masthan | AI Internship Project
Search
ENG
IN
23:26
16-05-2026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This AI system predicts suitable job roles based on user skills using Machine Learning and NLP techniques.
Paste your skills below
Enter Skills
python machine learning ai
Predict Job Role
Recommended Job Role: AI Engineer
Developed by Khadar Masthan | AI Internship Project
Search
ENG
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16-05-2026

## CONCLUSION

The AI Resume Screening System successfully predicts
suitable job roles based on user skills using Machine Learning techniques.
This project helped in understanding Python programming,
Machine Learning concepts, NLP basics, and Streamlit UI development.

### Future Enhancements

• Resume PDF Upload

• Database Integration

• Advanced AI Models

• Better Prediction Accuracy